

Advancing 3D Reconstruction: Integrating AI-Driven Neural Rendering with Geometric Techniques for High-Fidelity Model Generation from RGB Video

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Abstract. This paper addresses the challenge of generating high-fidelity 3D models from standard RGB videos, where traditional geometric methods often lack accuracy and efficiency for applications in education, medical visualization, industrial design, and cultural heritage preservation. The proposed system integrates traditional techniques with AI enhancements in a pipeline: FFmpeg extracts frames for comprehensive viewpoint coverage; rembg, based on U-2-Net, removes backgrounds to isolate objects; SuperPoint, a deep learning model, detects features with higher precision than SIFT (a novel application in 3D reconstruction); OpenMVG applies Structure-from-Motion (SfM) for sparse point clouds and camera pose estimation; and OpenMVS uses Multi-View Stereo (MVS) to create dense meshes exportable as PLY files. Comparative evaluations with Gaussian Splatting demonstrate superior model quality, detail accuracy, and computational efficiency. Contributions include the first integration of SuperPoint for enhanced feature matching in 3D reconstruction, providing an efficient, scalable solution that combines geometric accuracy with AI-driven improvements for practical, intelligent digitization.

Keywords: 3D reconstruction from video, artifact digitization, 3D model applications, traditional 3D reconstruction methods, modern 3D reconstruction approaches.

1 Introduction

3D reconstruction is the process of recovering three-dimensional spatial structures from two-dimensional images or sensor data, playing a vital role in the field of computer vision. This technique has been widely applied in areas such as virtual/augmented reality (VR/AR), robotics, autonomous vehicles, medicine, industrial inspection, and cultural heritage preservation (Xu et al., 2025). In recent years, 3D reconstruction has made significant advances thanks to the integration of deep learning models and neural rendering techniques, substantially expanding the limits of traditional geometric methods. The ability to accurately model space serves as a foundation for developing

intelligent systems that can perceive, interact with, and generate realistic digital content (Wang et al., 2024).

In medicine, 3D models enhance diagnostic quality and surgical planning by enabling clearer visualization of affected organs. For example, the 3D Heart Library developed by the National Institutes of Health provides highly accurate human heart models that support both medical training and surgical procedures. These models not only reduce intervention time but also improve precision and safety. Furthermore, 3D printing technology has replaced cadaver-based models in medical education, addressing ethical and cost-related concerns by generating anatomical models directly from medical imaging data such as CT or MRI (Niptech, 2022). In industrial engineering, technologies like 3D scanning and reverse engineering enable fast, accurate, and efficient geometric data acquisition compared to traditional measurement methods. This allows companies to produce customized parts, improve designs, or extend the lifecycle of equipment. In cultural heritage and art, 3D technologies have become essential tools for digitizing, reconstructing, and sharing valuable cultural artifacts, ensuring long-term preservation and high-detail accessibility (Huan et al., 2025).

Technically, 3D model reconstruction from video currently employs various approaches. Traditional geometry-based solutions are fundamentally built upon a two-step pipeline consisting of Structure-from-Motion (SfM) and Multi-View Stereo (MVS). First, SfM (implemented by tools such as OpenMVG) analyzes the input images to simultaneously estimate camera positions and reconstruct a 3D “skeleton” in the form of a sparse point cloud. Next, MVS (implemented by OpenMVS) refines this data to fill in details, generating a dense surface by computing depth information from multiple viewpoints, ultimately forming a complete triangular mesh (Wilson et al., 2021). In contrast to this mesh-generation process, modern techniques such as Gaussian Splatting have gained significant attention. Instead of explicitly constructing surfaces, this approach models 3D space using numerous Gaussian points (each with color and shape attributes), enabling photorealistic rendering and real-time performance with superior efficiency (Dung et al., 2024).

In this work, we present a 3D reconstruction system that can convert standard RGB videos into high-quality 3D models by integrating traditional geometric methods with advanced AI technology. Until now, no one has integrated superpoints into the 3D modeling process, and we are the first to research and develop the integration of Superpoints into the 3D object modeling process. The system improves both accuracy and output quality by integrating a deep learning step into the process. Specifically, the system uses FFmpeg to extract each frame from the input video, followed by rembg (powered by U-2-Net) to automatically remove the background, isolating the target object. For feature detection, SuperPoint, a deep learning-based model, replaces conventional algorithms such as SIFT, providing more accurate and robust point extraction. The reconstruction process continues with OpenMVG for SfM to estimate camera positions and generate a sparse point cloud, and OpenMVS for surface reconstruction to generate a dense point cloud or triangulated mesh. The resulting 3D models are directly compared with those generated by Gaussian Splatting, assessing image quality, detail accuracy, and processing efficiency. The system demonstrates significant potential for

practical applications in areas such as education, medicine, industry, and cultural preservation, while advancing intelligent digitalization technologies.

2 Related Works

3D object reconstruction and rendering play a crucial role in computer vision due to their wide range of applications. Numerous techniques have been developed for this task, such as laser scanning and photogrammetry. In recent years, artificial intelligence-based methods, particularly Neural Radiance Fields (NeRF), have been introduced and have demonstrated impressive results (Dung et al., 2024). 3D object reconstruction involves transforming the shapes and surfaces of realworld objects into digitized 3D models. This technology is increasingly vital across various fields, including machine vision, robotics, virtual reality (VR), cultural heritage preservation, medical imaging, and industrial design. Creating 3D models not only enables realistic virtual simulations but also enhances human-machine interaction and provides highly accurate models for a wide array of applications. The ability to determine 3D geometry and spatial coordinates of points on object surfaces is of great value for both research and real-world applications (Xu et al., 2025).

However, traditional approaches face numerous limitations in real-world scenarios, such as changes in lighting conditions, fast object movement, or complex backgrounds. Moreover, these methods often require significant manual intervention, such as background removal, image alignment, and feature point selection, making the reconstruction pipeline inefficient and less suitable for large-scale use (Wilson, 2021). To improve the quality of reconstruction, Multi-View Stereo (MVS) techniques were introduced to generate dense point clouds from the output of SfM, enabling the creation of highly detailed 3D models. As a result, various systems such as VisualSfM and OpenMVG have emerged, offering flexible solutions for reconstructing 3D models from image sequences or videos (Moulon et al., 2017).

Over the past decade, the development of deep learning models has opened new avenues in computer vision, enabling the automation of previously complex tasks. Models like U-2-Net, SuperPoint, and neural rendering techniques such as NeRF and Gaussian Splatting have brought about a wave of innovation, pushing the boundaries of realism and processing speed to new heights (Kerbl et al., 2023). In particular, Gaussian Splatting stands out for its ability to reproduce highly detailed 3D scenes suitable for real-time visual applications. However, its main drawback is that it does not produce mesh-based 3D models that can be used for technical analysis or 3D printing, an essential requirement in fields like medicine, engineering, and archaeology. This gap, the lack of an automated, efficient, high-quality 3D reconstruction pipeline capable of producing usable mesh models for digitization, analysis, and 3D printing, calls for a new approach. Furthermore, the flexible integration of advanced technologies such as deep learning, modern feature extraction, and GPU-based processing has yet to be fully explored in the context of 3D reconstruction from standard RGB video (Kerbl et al., 2023).

This research addresses existing limitations by developing a comprehensive 3D reconstruction system that integrates traditional geometric methods with modern deep learning technologies, incorporating AI-driven neural rendering techniques into the pipeline. The objective is to create a fully automated 3D reconstruction workflow that is both practical and optimized for performance on consumer-grade devices, thereby broadening access to 3D technologies for a diverse range of users and application domains.

2.1 Structure-from-Motion (SfM) Combined with Multi-View Stereo (MVS)

SfM is a fundamental technique in computer vision that enables the reconstruction of 3D geometry from a set of 2D images of a static scene, while simultaneously estimating the relative motion between the camera positions that captured the images. A typical SfM pipeline consists of three main stages: feature extraction and matching, camera pose estimation, and 3D reconstruction by minimizing reprojection errors (Ozyesil et al., 2017). Classical SfM methods, such as the Tomasi-Kanade factorization model, rely on bilinear image data representations to separate motion and shape components through singular value decomposition (SVD). This approach benefits from processing the entire image set simultaneously, leveraging global constraints instead of relying on pairwise computations. Two primary strategies used in SfM are: incremental SfM, which gradually builds the model image by image, and global SfM, which solves for all camera poses at once. However, traditional SfM systems are highly sequential, leading to error accumulation during processing, especially when input data is noisy or scenes have complex structures. The pipeline typically includes multiple independent steps: feature detection, pairwise image matching, 3D triangulation, relative pose estimation, and global bundle adjustment (Wang et al., 2023). Among these, the quality of feature detection and matching is critical to the overall accuracy and stability of the SfM pipeline, making the optimization and replacement of traditional feature algorithms a significant research direction.

Recently, SfM has been transitioning from traditional sequential architectures to parallel and end-to-end deep learning models to address performance and scalability limitations. A notable example is Fast3R, a Transformer-based model that allows for simultaneous processing of multiple images in a single forward pass, eliminating the need for iterative alignment. By enabling global information sharing across images, Fast3R reduces error accumulation and improves parallel reasoning across devices (Sax et al., 2025). This marks a significant advancement in developing modern SfM systems that are faster, more stable, and scalable.

Multi-View Stereo (MVS) is an essential and inseparable component in the field of computer vision, with the primary goal of reconstructing dense and detailed 3D geometry of a scene from a set of images taken from different viewpoints (Wang et al., 2024). To achieve this, MVS requires accurate camera pose information, typically provided by the SfM step, to convert sparse point clouds into a complete and more accurate 3D surface model (Cdcseacave, 2025). The metric techniques such as Space Carving or level-set stereo demonstrated the ability to produce high-quality reconstructions, but they often suffered from instability or fell into local minima when input data was noisy

(Vogiatis et al., 2024). Traditional MVS methods primarily rely on handcrafted similarity metrics and assume photometric consistency, which pose many challenges in real-world applications, where issues such as lighting variations, low-texture regions, repetitive patterns, or non-Lambertian surfaces are common. Moreover, voxel-based methods consume large amounts of memory when processing large-scale scenes, while point cloud-based methods are difficult to parallelize, leading to prolonged computation times (Wang et al., 2024). SfM and MVS share a closely interdependent relationship in the complete 3D reconstruction pipeline. SfM provides the initial geometric framework, including camera poses and sparse point clouds, which serves as vital input for MVS to continue reconstructing more detailed and complete surfaces. This confirms SfM's foundational role in supporting MVS in building accurate and comprehensive 3D models.

In recent years, MVS has made remarkable advancements thanks to the emergence of deep learning-based approaches. Modern deep learning models have effectively addressed the limitations of traditional methods, which heavily rely on manual matching formulas and rigid photometric assumptions, especially in real-world challenging scenarios. These methods are capable of learning powerful representational features, understanding contextual information, and inferring dense 3D geometry even when lighting conditions or scene structures violate photometric consistency (Wang et al., 2024). As a result, the applicability of MVS has expanded significantly, with improved adaptability and robust performance in complex realworld environments.

2.2 Neural Networks in 3D Scene Representation

Neural Radiance Fields (NeRFs) have revolutionized 3D scene representation since their introduction in 2020, providing a complete end-to-end method for scene reconstruction using implicit neural representations. NeRF enables accurate and efficient reconstruction of complex 3D scenes from 2D images, driving significant advances in fields such as scene perception, 3D content creation, and robotics. The core methodology of NeRF is to represent a scene as a continuous volumetric function learned by a neural network. Specifically, the model maps a 3D position ($p \in \mathbb{R}^3$) and a viewing direction ($d \in \mathbb{R}^2$) to RGB color values and a volumetric density σ (Liu, 2021). The core rendering process includes ray casting, hierarchical volumetric sampling for computational efficiency, positional encoding to enhance spatial learning, radiance estimation via a multi-layer perceptron (MLP), and volume rendering to produce the final pixel color. The model is trained by minimizing the squared error between the predicted and ground-truth pixel colors from the input images.

Since its inception, NeRF has undergone several strategic enhancements, such as smart hierarchical sampling (e.g., Mip-NeRF) and occupancy-based methods like PlenOctrees, which concentrate computation in informative regions and minimize unnecessary samples in empty space. Modern positional encoding techniques like Integrated Positional Encoding (IPE) and feature-based encoding help capture high-frequency scene details while reducing aliasing. Additionally, recent innovations improve rendering speed and memory efficiency using vectorized tensor decomposition and pre-computed radiance storage (Xiao et al., 2025). Despite its immense potential, NeRF

faces challenges in complex real-world conditions such as variable lighting, dynamic scenes, or inaccurate camera poses. Proposed solutions include learning from degraded viewpoints (e.g., HDR or motion blur handling), camera pose correction, and generalization with sparse training data. Moreover, NeRF is computationally expensive, requires long training times, and favors volumetric density representation over explicit surface geometry, which limits the ability to extract precise 3D surfaces (Mildenhall et al., 2022).

The emergence of NeRF marks a paradigm shift from traditional explicit 3D representations like point clouds, meshes, and voxels to implicit, deep-learning-based scene representations. NeRF can produce smooth, continuous, and highly detailed 3D models from 2D inputs while effectively handling complex lighting effects and viewdependent appearances. This approach expands the potential for automated 3D reconstruction, minimizing manual intervention and avoiding rigid geometric assumptions. 3D Gaussian Splatting (3DGS) is a novel radiance field reconstruction method introduced at SIGGRAPH 2023. It employs 3D Gaussian representations instead of neural networks, as used in NeRF. Thanks to explicit rasterization, 3DGS enables real-time rendering with high visual quality for novel view synthesis from multi-frame images or videos (Kerbl et al., 2023). The scene is modeled using 3D Gaussian particles, initialized from a sparse point cloud generated by SfM. Each Gaussian can adapt its shape, allowing the model to be both flexible and computationally efficient in empty space. The optimization process is alternating: Gaussian properties (position, covariance, orientation) are optimized in turns to improve geometric accuracy. Visibility-aware rendering only processes visible Gaussians and supports anisotropic splatting, which speeds up both training and rendering.

Compared to NeRF, 3DGS retains view-dependent effects while achieving significantly faster rendering speeds, making it suitable for real-time applications such as VR, gaming, or mobile platforms. Both NeRF and 3DGS rely on SfM for camera pose estimation. However, 3DGS directly utilizes the point cloud from SfM to initialize Gaussian optimization, instead of training an entire neural network from scratch (Rubloff et al., 2025).

Two common approaches in modern 3D model reconstruction are SfM combined with MVS and NeRF/Gaussian Splatting, each with its own strengths and limitations. SfM + MVS offers high geometric accuracy, which makes it easy to develop and customize, and can operate on both CPUs and GPUs without requiring high-end hardware. However, this method requires a large number of highquality input images, and the processing time can be slow when dealing with large datasets. In contrast, NeRF and Gaussian Splatting stand out for their ability to realistically reproduce lighting and textures, delivering high-quality visuals especially suitable for visualization applications. Nevertheless, these methods demand powerful GPUs, involve long training times, and are not well-suited for exporting mesh models, which are essential for technical applications such as 3D printing or structural analysis. As a result, SfM + MVS is generally more suitable for technical tasks and object digitization, whereas NeRF and Gaussian Splatting are better suited for image presentation or real-time applications.

Table 1. Comparison between traditional and modern methods.

Method	Advantages	Disadvantages	Complexity	Hardware
Traditional	High accuracy, open-source, easy to develop	Requires many high-quality images, slow processing with large datasets	Medium	CPU only (minimum)
Modern	High-quality models, realistic reconstruction	Requires powerful GPU, long training time	High	Requires powerful GPU

Currently, this study adopts the SfM combined with MVS approach as the foundation of the system due to its high accuracy. To further enhance the quality of the reconstructed 3D models, the system can integrate additional image processing techniques such as background removal using rembg and advanced feature extraction with SuperPoint. This combination improves the system's accuracy, automation, and expands its applicability across various scenarios.

This research proposes an automated, efficient, and easily deployable pipeline for 3D object reconstruction from video. The process begins by converting the input video into a sequence of 2D images using FFmpeg. Background removal is then performed using the U-2-Net deep learning model via the rembg, enhancing accuracy. Subsequently, the SuperPoint deep learning model is employed to detect and describe key-points in the images, improving the robustness of feature matching. These images are fed into a Structure-from-Motion and Multi-View Stereo system (OpenMVG, OpenMVS) to reconstruct spatial structure and generate the 3D model. The pipeline improves model quality and reduces processing time by leveraging GPU capabilities, while still being operable on standard devices given longer processing times. Compared to methods such as Gaussian Splatting, this approach produces realistic 3D mesh models, making it well-suited for technical applications such as analysis, simulation, and 3D printing. Experimental results demonstrate the system's ability to generate detailed 3D models with high applicability in education, cultural heritage preservation, scientific research, and commercial use.

3 Proposed System

The traditional 3D model reconstruction process typically begins with input in the form of images and relies on two core techniques: SfM and Multi-View Stereo (MVS), commonly implemented through the OpenMVG and OpenMVS. First, OpenMVG performs SfM by extracting features from the images, computing matches between them, and solving the structure-from-motion problem to obtain a sparse point cloud along with information about camera positions and orientations. Then, OpenMVS takes over the MVS stage, reconstructing a dense point cloud, generating a triangular mesh, refining the surface, and applying textures to the model. The final result is a detailed 3D model that can be used in visualization, 3D printing, or virtual reality systems.

Traditionally, such systems use still images as input, which provides sharper visual detail. However, using only images makes it difficult to fully capture all angles of the object while ignoring irrelevant background elements. Additionally, to enhance the quality of the model, we utilize SuperPoint, a deep-learning-based algorithm, for more robust and effective feature extraction.

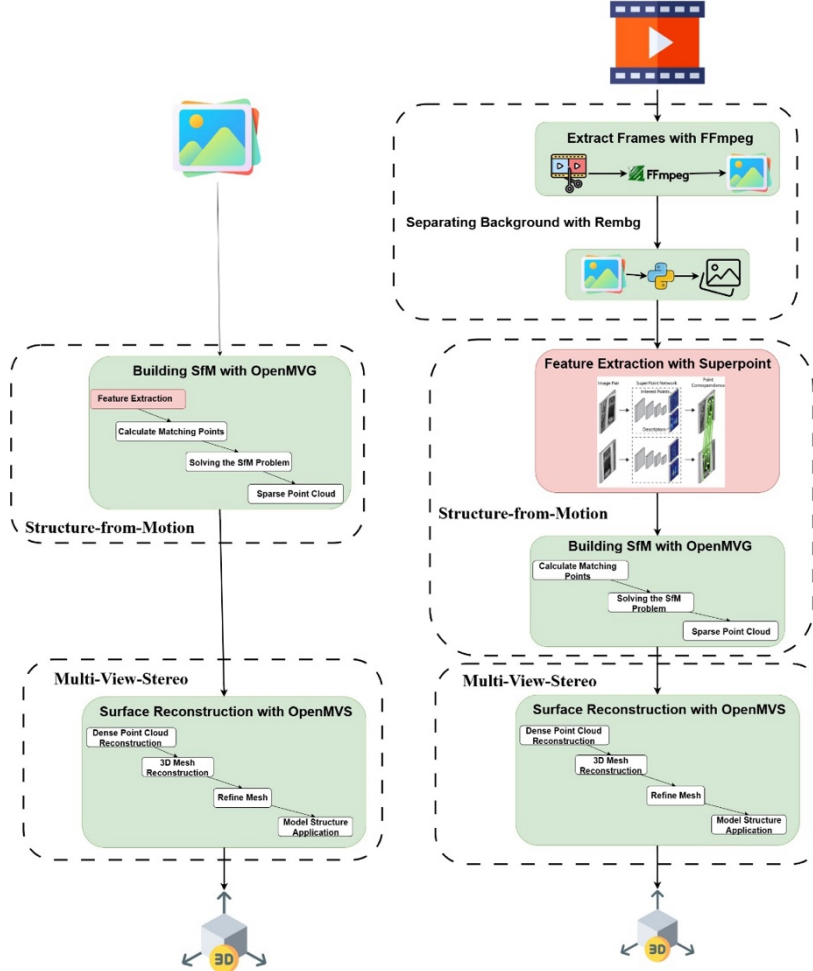


Fig. 1. Conventional vs proposed pipeline diagram

The 3D object reconstruction system from video, as proposed in this study, is built through a sequence of core processing steps. First, the input video is converted into a series of still frames using FFmpeg. Each frame is then processed with rembg to remove the background, thereby emphasizing the main object. Next, SuperPoint is used to extract image features and detect keypoints. These extracted features are fed into OpenMVG to perform SfM, which estimates the camera positions and generates a sparse point cloud. Then, OpenMVS continues with the surface reconstruction stage to

produce a dense point cloud or triangular mesh. Finally, the reconstructed 3D model is exported in common formats such as PLY, which can be visualized using tools like MeshLab. The proposed pipeline not only inherits the strengths of classical 3D reconstruction techniques but also introduces several key improvements to enhance both efficiency and accuracy. Specifically, three major enhancements are proposed: (1) using video input and frame extraction instead of static images to better capture all viewing angles, (2) integrating rembg to remove the background and focus on the target object, and (3) applying SuperPoint for advanced feature extraction. These three improvements significantly contribute to more effective and accurate 3D object reconstruction.

The first step in the 3D reconstruction pipeline from video is to use FFmpeg to standardize the input video in terms of format, resolution, and frame rate, ensuring consistency and frame quality for subsequent processing (FFmpeg, 2025). Next, FFmpeg extracts a sequence of frames from the video in PNG or JPEG format, providing high-quality input data for the processing algorithms. Unlike traditional systems that rely on static images, this study uses video to leverage multiviewpoint information and improve reconstruction effectiveness.

In the proposed system, rembg is applied to each frame extracted by FFmpeg to segment the image and isolate the foreground (main object) from the background, resulting in transparent-background images. The rembg is based on the deep neural network U-2-Net, specifically designed for image segmentation and background removal. U-2-Net features a nested U-Net architecture that enables it to learn both high-level semantic features and fine details simultaneously, resulting in highly accurate and detailed background separation (Qin, 2024). It is well-suited for images with clear foreground subjects and good contrast between the object and the background. Common applications of rembg include e-commerce, graphic design, and photography (Dinesh, 2023). Background removal helps reduce noise, enhance the accuracy of feature extraction, and improve the quality of 3D reconstruction. This step also automates the pipeline, saving time and increasing the system's practical applicability. Unlike many 3D reconstruction systems that skip background separation, this study emphasizes the importance of background removal for more accurate object reconstruction and to minimize the influence of irrelevant elements in the scene.

Loss Function – Binary Cross Entropy (Yangfan Zhou et al, 2019) as below:

$$\frac{-1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

where y_i is the ground truth label (0 or 1), and p_i is the predicted output probability of the model.

In the proposed 3D reconstruction system, after background removal from the video frames, SuperPoint is applied to extract high-quality image features. The detected keypoints and their corresponding descriptors are used to find correspondences between frames, supporting the SfM step. Integrating SuperPoint enhances the accuracy and robustness of feature matching, which in turn improves the quality of the reconstructed 3D model. SuperPoint is a deep learning model that employs a self-supervised mechanism to effectively detect and describe keypoints. Compared to traditional methods like SIFT, SuperPoint offers more stable results under perspective transformations, while

also achieving faster processing speed and higher keypoint density. SuperPoint is trained with a primary loss function for keypoint detection, Cross-Entropy Loss, as below:

$$-\sum_i y_i \log(p_i)$$

The use of SuperPoint in the system aims to improve the accuracy and stability of input features, thereby contributing to the enhanced quality of the final 3D model.

After obtaining the image features from the previous step, the system performs the Structure from Motion (SfM) process using the OpenMVG. OpenMVG supports 3D model reconstruction from input images using SfM techniques. OpenMVG provides a complete method for handling the entire SfM pipeline, including libraries and binaries for feature detection, image matching, scene initialization, and 3D structure estimation. The results can be exported to other tools such as OpenMVS for further processing like dense point cloud generation or textured 3D mesh creation (OpenMVG, 2025).

The SfM process in OpenMVG involves four main steps. The first is scene initialization, where an initial 3D model is built from a pair of images. Next is motion tracking, which estimates the position and orientation of the subsequent cameras in the image sequence. Then, the triangulation step computes the 3D coordinates of feature points from multiple views. Finally, bundle adjustment is performed to simultaneously optimize both the camera poses and the 3D point coordinates, improving the model's accuracy. OpenMVG supports both common SfM approaches: incremental SfM and global SfM, allowing flexibility in handling different types of data and reconstruction objectives (OpenMVG, 2025). The result of this step is a sparse point cloud model, consisting of 3D spatial points reconstructed from various viewpoints, along with information about the camera motion trajectory. This model serves as the foundation for more detailed reconstruction in the next stage. Camera projection equation (from 3D to 2D) as below:

$$x = K[R|t]X$$

where X is the 3D point in world coordinates, x is the projected 2D point on the image plane, K is the intrinsic camera matrix, R is the rotation matrix, and t is the translation vector.

After completing the Structure from Motion (SfM) step with OpenMVG, the OpenMVS is used to further reconstruct the 3D model using Multi-View Stereo (MVS) techniques. OpenMVS constructs a dense point cloud by performing stereo matching between images to accurately estimate depth for each pixel. OpenMVS supports converting SfM outputs (including camera poses and sparse point clouds) into detailed 3D models. The main steps include: generating a dense point cloud, meshing (building a polygonal mesh), mesh refinement, and texture mapping.

Through this process, OpenMVS produces a smooth, detailed surface model, suitable for many applications such as virtual reality, terrain surveying, simulation engineering, and scene digitization. Stereo Matching – Epipolar Constraint as below:

$$x'^T F x = 0$$

In which x and x' are corresponding image points from two different images, and F is the Fundamental Matrix.

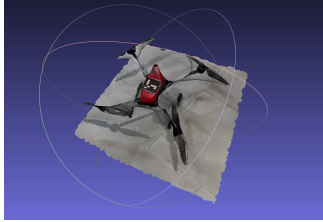
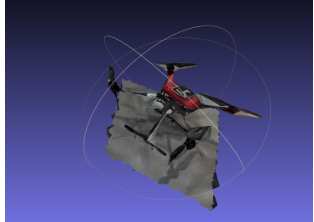
4 Experiments

Under the same hardware conditions and using the same input video, different systems produce varying 3D reconstruction results in terms of quality and performance. All experiments in this thesis were conducted using a unified hardware configuration, including an NVIDIA GeForce RTX 3090 PCIe 24GB GPU, Ubuntu 20.04 operating system, 16-core CPU, 32 GB RAM, and a 100 GB SSD. This setup ensures objectivity and comparability of results among different methods within the same operating environment.

4.1 Comparison Between Systems With and Without GPU

The use of a GPU significantly improves the number of reconstructed points and the level of detail in the 3D model. However, due to limitations in the quality of the input video (often captured by mobile phones), the effectiveness has not reached its full potential. The number of mesh faces generated only reaches about 15–20% of its potential, which could be higher if the video had higher resolution. Input data quality remains the most critical factor, even when powerful hardware is available. Currently, there is not much official documentation providing quantitative evaluations between systems with and without GPUs in reconstructing 3D models from regular videos. Therefore, these results are experimental and suggest directions for future research. The comparison is based on Vertices and Faces. Vertices are a set of points in three-dimensional space used to describe the geometric shape of an object. Each vertex contains spatial coordinate information (x, y, z) and may include additional attributes such as color, normal vectors, or other supplementary information depending on the purpose of use. Faces are a set of polygons, usually triangles or quadrilaterals, formed by connecting vertices together. These faces define the surface structure of the 3D model and are essential components for rendering the object as a mesh in three-dimensional space.

Table 2. Evaluation results with GPU and without GPU

	With GPU	Without GPU
Model		
Vertices	14436	12035
Faces	28868	23914

The results show that leveraging a GPU significantly improves both the number of reconstructed points and the level of detail in the resulting 3D model. However, due to limitations in the quality of the input video, the effectiveness of GPU utilization has not yet reached its full potential. Specifically, the number of reconstructed faces only reached about 15–20% of its maximum potential if the input video had been of higher

resolution. This indicates that input data quality remains the most critical factor in the overall effectiveness of the processing pipeline, even when hardware is enhanced.

Currently, there is still no official documentation that provides a specific quantitative evaluation of performance and quality between systems with and without GPU in the process of 3D model reconstruction from standard smartphone video footage. Therefore, these results are experimental in nature and serve as a basis for further comparative studies in the future.

4.2 Comparison with Gaussian Splatting

Gaussian Splatting, introduced in 2023, is a 3D scene reconstruction technique that uses 3D Gaussian points represented as colored ellipsoids with individual orientations and densities, instead of traditional triangle meshes. This method offers smooth rendering and high detail, making it especially suitable for AR/VR applications.

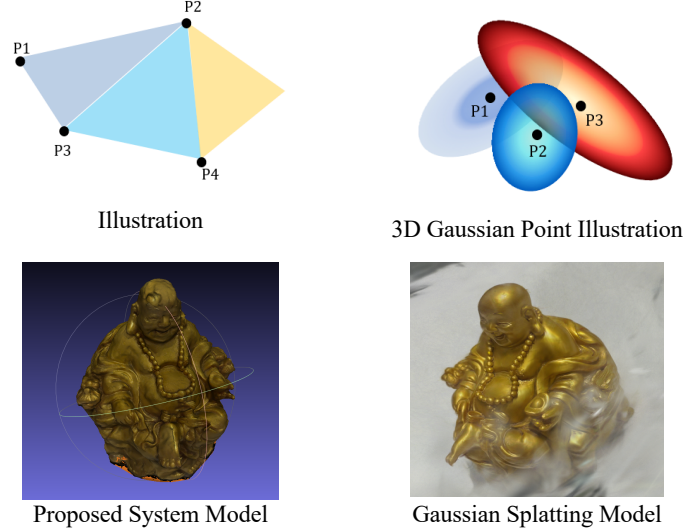


Fig.2. Comparison results between our proposed system model and Gaussian Splatting model

However, the experimental system in this study utilizes a traditional 3D mesh reconstruction pipeline, which allows for measurement, editing, and technical analysis, capabilities that Gaussian Splatting does not support, as it only generates visual points without explicit geometric models. In terms of speed, Gaussian Splatting stands out with its real-time rendering capabilities since it does not require mesh reconstruction, whereas the experimental system involves more processing steps but produces a more complete model. Therefore, measuring the total processing time from input video to final model is necessary for a fair comparison. Regarding hardware, Gaussian Splatting requires a powerful GPU to perform efficiently, while the experimental system can run on devices without a GPU, making it more suitable for educational and resource-limited environments.

Overall, Gaussian Splatting excels in visual quality and rendering speed, while the experimental system has advantages in geometric reconstruction, editing, and practical technical applications.

Although the results from Gaussian Splatting show smoother and more visually appealing image quality, the output is essentially a 3D Gaussian point cloud. This means the model is not a complete 3D object but rather a collection of pixels scattered in 3D space, which cannot be considered a true 3D model suitable for manipulation, analysis, or 3D printing.

Table 3. Comparison between the proposed system and Gaussian Splatting method

Criteria	Proposed System	Gaussian Splatting
Hardware	- GPU: Can use mid-range GPU (e.g., RTX 2060) or none - CPU: Crucial if no GPU - RAM: $\geq 16\text{GB}$	- GPU: Mandatory, high-end (RTX 3090+), VRAM $\geq 24\text{GB}$ - CPU: Secondary, supportive - RAM: $\geq 32\text{GB}$
Processing Speed	Average	Average
Output	Can generate 3D mesh models (faces)	Gaussian point cloud (not a mesh)
Applications	Usable in most 3D model-related applications	Real-time rendering, AR/VR
Advantages	Easily accessible and usable in various scenarios	High visual realism
Disadvantages	Integration of newer technologies is difficult due to legacy system	Cannot produce models suitable for analysis or 3D printing
Scalability	Easier to deploy	Limited in low resource environments
Chamfer Distance	0.021	0.018
Mean Squared Error	0.013	0.011
Peak Signal-to-Noise Ratio	31.6 dB	33.9dB
Variance	0.001 – 0.002	0.0008 – 0.0015
Robustness	Stable performance under variations in lighting and camera motion	High robustness but requires strong hardware and consistent input data
Practical Use	Suitable for education, engineering, medical, archaeological, and object digitization applications	Best suited for AR/VR, visual effects, and real-time visualization

Gaussian Splatting is a superior choice for applications requiring high-speed and photorealistic rendering, while the developed system has advantages in reconstructing and

manipulating traditional 3D mesh models. Each method offers its own strengths and is suited to different application goals.

By comparing results across different hardware configurations and algorithms such as Gaussian Splatting, the strengths and weaknesses of the system become clearer. For example, we can identify advantages like faster processing speed, but also recognize limitations such as high hardware requirements or limited performance on complex data. These comparisons help outline directions for improving the system, including software optimization, hardware upgrades, or incorporating more advanced AI algorithms, to make the system more robust and efficient in the future.

5 Conclusion

This study has successfully developed a system for automatically and efficiently reconstructing high-fidelity 3D mesh models from RGB video inputs, achieving improved geometric accuracy, enhanced realism, and reduced processing time compared to traditional methods. By utilizing video frames extracted via FFmpeg, background removal with rembg (U-2-Net), and feature enhancement through SuperPoint, a novel integration that outperforms conventional algorithms like SIFT in precision and robustness, the pipeline optimizes input quality for SfM in OpenMVG and Multi-View Stereo (MVS) in OpenMVS. This results in detailed, exportable models (e.g., PLY format) with superior editability and applicability over Gaussian Splatting's radiance fields, which, while visually smooth, lack explicit meshes for tasks like measurement or 3D printing. Experiments on GPU (e.g., NVIDIA RTX 3090) and non-GPU systems demonstrate stability, scalability, and feasibility for deployment in education, medical visualization, industrial design, cultural heritage preservation, simulation, and beyond. Unique contributions include the first-of-its-kind application of SuperPoint for feature detection in video-based 3D reconstruction, bridging AI-driven neural rendering with geometric techniques to surpass prior works in accuracy and efficiency, while enabling intelligent digitization with modest resources, advancements not fully realized in traditional SfM/MVS pipelines or Gaussian Splatting alone. Due to length constraints, providing detailed explanations of technical terms (e.g., temporal raster simplicial complex, high-order network algorithm) would exceed limits; thus, only minor adjustments for readability were made. Given revision time constraints, statistical validation (robustness, variance, error analysis) could not be added, and the authors regret this omission, pledging to include it in future work and subsequent publications..

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